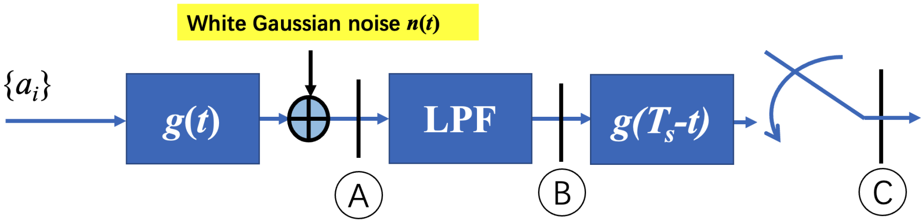
Communications and Networks (Spring 2022)

Problem Set 4

Issued: Apr. 12 Due: Apr. 24

Readings: Haykin - Communication System 8.3, 9.1-9.5

1. In this problem, we investigate the SNR (Signal-to-Noise Ratios) at different points in a baseband transmission system with the matched filter and assume no-ISI.



The baseband waveform is a square-root raised cosine with roll-off factor $\alpha \in [0, 1]$. At the receiver, the signal is sampled at T_s after the matched filter. The symbols $\{a_i\}$ take values in $\{0, 1, 2, 3\}$ with the equal probability. The double-sided PSD (Power Spectrum Density) of the noise is $n_0/2$.

a) With the no-ISI condition, what is the bandwidth W of $G(f)$? Write the explicit form of the frequency representation $G(f)$.

b) Derive the SNR at point B, from the baseband waveform perspective. You can keep the integral term of $G(f)$.
(Hint: The distribution of $\{a_i\}$ should be considered.)

c) Derive the SNR at point C, after sampling at T_s. Compare the SNR at point A, B, C and explain.

2. In a baseband transmission system, the transmitted signal is

$$x = \begin{cases} 0, & \text{symbol '0' is transmitted} \\ A, & \text{symbol '1' is transmitted} \end{cases},$$

and the received signal is $y = x + n$, where n is the equivalent additive noise, following the Laplace distribution

$$f(n) = \frac{1}{2\lambda} e^{-\frac{|n|}{\lambda}}$$

a) Assume that the transmitted symbols are equiprobable. Determine the detection threshold with minimum SEP (Symbol Error Probability) and calculate the corresponding SEP.

b) If the detection rule at the receiver is biased, specifically

$$\tilde{x} = \begin{cases} \text{'0'}, & y < \frac{1}{4}A \\ \text{'1'}, & y \geq \frac{1}{4}A \end{cases}.$$

Find the conditional symbol error probability $P(\tilde{x} = \text{'0'}|x = \text{'1'})$ and $P(\tilde{x} = \text{'1'}|x = \text{'0'})$.

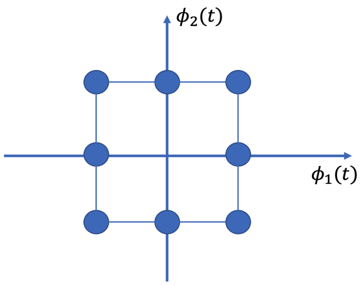
c) Find the prior distribution of the transmitted symbols so that the detection rule in b) minimizes the SEP for the given prior distribution.
(Hint: the question is different from minimizing the SEP given the detection rule)

3. In a baseband transmission system with binary antipodal symbol set, the data rate is 5kbps and the double-sided PSD (Power Spectrum Density) of the additive Gaussian noise is $n_0/2$, where $n_0 = 10^{-9}$W/Hz.

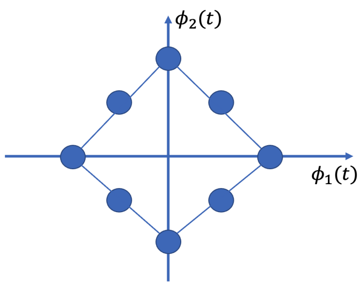
a) If the system is required to be reliable, producing less than 100 bit errors on average in one day, what is the required SEP (Symbol Error Probability) of the system?

b) Determine the minimum energy per symbol E_s and the average transmiting power P_t.
(Hint: use qfuncinv$(\cdot)$ in Matlab to calculate)

4. Below are two schemes of 8-symbol constellation. Explain which one is better, in terms of SEP (Symbol Error Probability), given the same power constraint.



Scheme 1



Scheme 2

5. In this problem, we compare different methods in passband modulation. Assume the double-sided PSD of noise is $n_0/2$ and all the symbols are equiprobable.

a) Given the energy per symbol E_s, find the SEP (Symbol Error Probability) of 4QAM and 16QAM. Explain why the SEP is lower for 4QAM.

b) Given the energy per bit E_b, find the SEP of 4QAM and 16QAM. Which one has a lower SEP?

(Hint: it is recommended to draw the SEP-SNR curves using Matlab)

c) Given the energy per bit E_b, compare the SEP of 16QAM and 16PSK. Explain intuitively why QAM is better?